SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

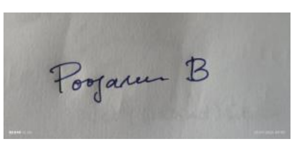
QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I POOJASREE B(192411091) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Annexure A

ARTICLE REVIEW

<u>INSTRUCTIONS</u>

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

(i) Article Citation Details

Authors: Younghak Shin, Hemin Ali Qadir, Lars Aabakken, Jacob Bergsland, and Ilangko Balasingham.

Title: *Automatic Colon Polyp Detection Using Region Based Deep CNN and Post Learning Approaches*

Journal: *IEEE Access*

Volume: 6

Pages: 40950–40962

Year: 2018

DOI: 10.1109/ACCESS.2018.2856402

(ii) PDF / Valid DOI Link

DOI:
https://doi.org/10.1109/ACCESS.2018.2856402

PDF:
Article PDF

The PDF you uploaded is the same article, titled **"Automatic Colon Polyp Detection Using Region-Based Deep CNN and Post Learning Approaches."**

(iii) Article Review

Task 1: Article Summary

(a) Full Citation

Shin, Y., Qadir, H. A., Aabakken, L., Bergsland, J., & Balasingham, I. (2018). *Automatic Colon Polyp Detection Using Region Based Deep CNN and Post Learning Approaches.* **IEEE Access, 6, 40950–40962.** DOI: **10.1109/ACCESS.2018.2856402**.

DOI link: https://doi.org/10.1109/ACCESS.2018.2856402

Domain/Application Area: Medical image processing, computer vision, and healthcare.

Specific ML Problem: Automatic detection and localization of colorectal polyps in colonoscopy images and videos using deep learning. The problem is challenging because polyps vary significantly in shape, size, texture, and color, while normal colon structures can appear similar to polyps.

(b) Key Objective and Research Gap

The main objective of the article is to develop a reliable deep-learning system that can automatically detect and localize colorectal polyps in colonoscopy images and videos. The authors identified that existing polyp-detection methods often depended on hand-crafted features such as color, texture, shape, LBP, and edge information. These features can be confused by normal structures that resemble polyps.

Another important research gap was the limited availability of labeled colonoscopy images, which makes training very deep CNN models difficult. At the time of the study, region-based CNN methods such as Faster R-CNN had achieved strong results in general object detection, but had not yet been applied to automatic colon-polyp detection.

To address these gaps, the authors introduced a Faster R-CNN-based detector using Inception-ResNet-v2 through transfer learning. They also investigated domain-specific image augmentation to overcome the small training dataset. Finally, they proposed automatic false-positive learning and video-specific offline learning to reduce polyp-like false detections and improve performance in colonoscopy videos.

Task 2: Methodology Analysis

(a) ML Techniques and Dataset

ML Techniques Used

The main technique is **Faster R-CNN**, a region-based deep convolutional neural network. It contains:

1. **Region Proposal Network (RPN):** Generates candidate rectangular regions that may contain polyps.
2. **Fast R-CNN detector:** Classifies the proposed regions as polyp/background and refines their bounding-box coordinates.
3. **Inception-ResNet-v2:** Used as the deep CNN feature extractor through transfer learning.
4. **Image augmentation:** Rotation, flipping, shearing, zooming, blurring, brightening, and darkening are used to increase training variation.
5. **False Positive (FP) Learning:** Automatically collects strong polyp-like false detections from normal videos and retrains the detector.
6. **Offline Learning:** Uses reliable detections from a particular test video to retrain the detector specifically for that video.

The Inception-ResNet-v2 model was pretrained using the MS COCO dataset containing 112K images and 90 object categories. Its pretrained weights were then fine-tuned for polyp detection.

Dataset

Dataset	Size	Purpose	Details
CVC-CLINIC	612 images	Training	388×284 pixels; 31 videos; 31 unique polyps
ETIS-LARIB	196 images	Still-image testing	1225×966 pixels; 34 videos; 44 polyps; 208 total polyp instances
ASU-Mayo	20 videos used	Video evaluation	10 positive + 10 negative videos; 5 negative videos used for FP learning
CVC-ClinicVideoDB	18 videos	Video testing	11,954 frames; 10,025 contain polyps; 768×576 pixels

The CVC-CLINIC images were used for training and ETIS-LARIB for evaluation. Ground-truth polyp regions were annotated by skilled endoscopists.

The CVC-CLINIC images were expanded to **18,594 images using Augmentation-I** and **28,600 images using Augmentation-II**.

Training Preparation

The RPN uses IoU values of **0.6 for positive samples and 0.3 for negative samples**. Non-maximum suppression is used to remove highly overlapping proposals. Training uses SGD with momentum 0.9, learning rate 1e-3, and a maximum of 30 epochs.

(b) Mapping to CO 4 Topics

CO 4 Topic	Connection
Inductive Learning	The model learns visual patterns of polyps from labeled training images and generalizes them to unseen colonoscopy images.
Decision Trees	Not directly used in this article.
Statistical Learning	The model learns statistical representations/features from image data and uses classification probabilities for polyp/background detection.
Reinforcement Learning	Not used; there is no reward-action-state learning process.

Most relevant CO 4 concept: Inductive Learning, because the detector learns a general polyp-detection function from examples and applies it to unseen images/videos.

Methodological Strength

A major strength is the combination of **transfer learning, domain-specific augmentation, and post-learning**. Transfer learning allows a very deep CNN to be trained despite having only 612 original training images. The FP-learning method also directly addresses the clinically important problem of false alarms.

Methodological Limitation

The original training dataset is relatively small and contains mainly positive polyp images. Consequently, the detector can generate many false positives when encountering normal structures that resemble polyps. The authors themselves report that this caused a substantial reduction in precision on video data.

Task 3: Results & Critical Evaluation

(a) Key Results

Still-Image Results on ETIS-LARIB

Training Method	TP	FP	FN	Precision (%)	Recall (%)	F1 (%)	F2 (%)
Without Augmentation	82	89	126	48.0	39.4	43.3	40.9
Rot-Augmentation	147	99	61	59.8	70.7	64.8	68.2
Augmentation-I	**167**	26	41	**86.5**	**80.3**	**83.3**	**81.5**
Augmentation-II	148	14	60	91.4	71.2	80.0	74.5

Augmentation-I gave the best balance of precision, recall, F1 and F2, even though Augmentation-II generated more training images.

Comparison with Previous Methods

Method	TP	FP	FN	Precision (%)	Recall (%)	F1 (%)	F2 (%)
CUMED	144	55	64	72.3	69.2	70.7	69.8
OUS	131	57	77	69.7	63.0	66.1	64.2
UNS-UCLAN	110	226	98	32.7	52.8	40.4	47.1
CUMED + OUS	159	38	49	80.7	76.4	78.5	77.2
Proposed Aug-I	**167**	**26**	**41**	**86.5**	**80.3**	**83.3**	**81.5**
Proposed Aug-II	148	14	60	91.4	71.2	80.0	74.5

The proposed Aug-I model detected **167 of 208 polyps** and outperformed the compared methods in the reported precision, recall, F1 and F2 measures.

Video Results

For the 10 positive ASU-Mayo videos, Aug-I achieved **81.4% recall and 73.3% precision**. FP learning increased precision to **88%**, while offline learning achieved **84.2% recall and 83.4% F1**. All three approaches achieved a **100% Polyp Detection Rate (PDR)**.

For the five negative ASU-Mayo videos, automatic FP learning increased specificity from **71.1% to 97.7%**, an improvement of 26.6 percentage points.

For CVC-ClinicVideoDB, Aug-I achieved **83% precision, 80.2% recall, 81.6% F1 and 80.7% F2**. FP learning produced the highest precision of **92.2%**, while offline learning produced **84.3% recall and 86.9% F1**.

The reported mean processing time was approximately **0.39 seconds per frame (390 ms)** using an NVIDIA GTX1080 GPU.

(b) Critical Evaluation

The reported results are promising, particularly the **86.5% precision and 80.3% recall** on the still-image test set and the improvements obtained through FP and offline learning. However, these results should not be interpreted as clinical-level accuracy.

Potential Biases and Limitations

1. **Small original training dataset:** Only 612 CVC-CLINIC images were available for training.
2. **Dataset bias:** The images come from particular colonoscopy datasets and may not represent different hospitals, cameras, patient populations, or imaging conditions.
3. **Positive-sample bias:** The detector was primarily trained using polyp images, making it harder to learn the appearance of normal structures.
4. **Annotation issues:** Some detections considered false positives were actually polyps that had not been annotated by experts in particular frames.
5. **Limited external validation:** The authors used public datasets, but broader multi-center validation would be necessary before clinical deployment.
6. **No AUC-ROC:** AUC-ROC is not reported because the study focuses on object detection metrics such as precision, recall, F1, F2, PDR and processing time.

Additional Experiments That Would Strengthen the Study

- Test the model on a **completely independent multi-hospital dataset**.
- Compare Faster R-CNN against newer detectors such as YOLO and modern transformer-based detectors.
- Perform **cross-validation across hospitals or datasets**.
- Evaluate performance separately for small, medium and large polyps.

- Measure sensitivity at clinically relevant false-positive rates.
- Conduct a prospective clinical study comparing AI-assisted colonoscopy with conventional colonoscopy.
- Evaluate the system on different endoscopy cameras and image qualities.

(c) Real-World Applicability

The system could be deployed as a **computer-aided detection (CADe) tool during colonoscopy** to highlight suspicious polyp regions and help endoscopists avoid missing lesions. It could also be useful for offline analysis of recorded colonoscopy or wireless capsule endoscopy videos.

However, industrial deployment faces several challenges:

- **Data availability:** Large, diverse, accurately annotated medical datasets are difficult to obtain.
- **Hardware:** Inception-ResNet/Faster R-CNN requires substantial computational resources.
- **Real-time performance:** 390 ms per frame is approximately 2.6 frames/second, which is slower than standard real-time colonoscopy video.
- **Generalization:** Models trained on one hospital's equipment may not perform equally well elsewhere.
- **False positives:** Excessive alerts could distract clinicians.
- **Ethical and regulatory concerns:** Clinical AI requires validation, patient privacy protection, regulatory approval, and appropriate human oversight.

The authors themselves identify processing speed as a disadvantage if real-time detection is required.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

1. Inductive Learning from Limited Examples

I learned that inductive learning can generalize from a relatively small set of labeled examples when a pretrained model is used. The researchers used **transfer learning from Inception-ResNet-v2** instead of training the deep network from scratch. Training from scratch produced only **33.7% recall and 27.1% precision**, showing the importance of prior learned knowledge.

CO 4 connection: Inductive Learning.

2. Importance of Data Preparation in ML

The study showed that simply increasing the number of training images does not guarantee better performance. Augmentation-I produced **18,594 images** and performed better in recall and F1 than Augmentation-II with **28,600 images**. This demonstrates that the quality and relevance of training examples are more important than quantity alone.

CO 4 connection: Statistical Learning and model generalization.

3. Learning from Errors Improves the Model

The false-positive learning method demonstrates how difficult negative examples can be used to improve a trained classifier. The system collected polyp-like false detections from normal videos and retrained the detector. This improved specificity from **71.1% to 97.7%** on the reported negative-video evaluation.

CO 4 connection: Inductive Learning and error-driven model improvement.

(b) Proposed Novel Experiment / Enhancement

Proposed Experiment: Multi-Dataset Fine-Tuning with a Lightweight Real-Time Detector

I would extend the research by developing a **lightweight real-time polyp detector** and training it using multiple independent colonoscopy datasets. Instead of relying mainly on Faster R-CNN with Inception-ResNet-v2, a modern lightweight object-detection architecture could be fine-tuned using combined datasets containing different resolutions, cameras, lighting conditions, and patient cases.

The experiment would compare the original Faster R-CNN model with the lightweight model using **precision, recall, F1-score, false positives per frame, detection latency and GPU memory usage**. The model could also retain the paper's FP-learning strategy to improve its handling of polyp-like normal structures.

This experiment is feasible because the original study already demonstrates that multiple public datasets can be used for training and evaluation. The expected impact would be a better balance between **detection accuracy and real-time processing**, addressing one of the main limitations identified by the authors: the approximately **390 ms processing time per frame**.

Overall Review

The article presents a strong application of **inductive deep learning to medical image detection**. Its main contribution is not only the use of Faster R-CNN with Inception-ResNet, but also the combination of **transfer learning, domain-specific augmentation, false-positive learning and offline learning**. The reported results demonstrate substantial improvement over the compared methods. However, the relatively small and dataset-specific training data, potential annotation issues, and processing speed limit direct clinical deployment. Overall, the paper provides a useful example of how machine-learning models can be adapted to difficult real-world problems when data are limited and false positives are important.